

PROJECT REPORT ON

Handwritten Digit Recognition using OCR Systems

Course Name: Deep Learning

Submitted by:

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ABSTRACT

This microproject presents a deep learning-based Optical Character Recognition (OCR) system for handwritten digit recognition. Leveraging Convolutional Neural Networks (CNNs) and the MNIST dataset, the model is trained to classify digits (0–9) with high accuracy. The system is deployed via a Streamlit web interface, enabling real-time predictions and interactive evaluation. Key preprocessing steps include grayscale normalization and input reshaping, while the model architecture features convolutional layers, ReLU activations, max pooling, dropout, and a softmax output. Evaluation metrics such as accuracy, precision, recall, and F1-score demonstrate the model's robustness and generalization capability. This project highlights the practical application of deep learning in OCR tasks and provides a scalable foundation for future extensions in document digitization and intelligent automation.

INTRODUCTION

Optical Character Recognition (OCR) is a transformative technology that converts handwritten or printed text into machine-readable formats. In domains such as banking, education, and government digitization, OCR systems play a critical role in automating data entry and improving accessibility. This project focuses on digit-level recognition using deep learning, specifically Convolutional Neural Networks (CNNs), to classify handwritten digits from the MNIST dataset. Traditional OCR systems often rely on handcrafted features and rule-based logic, which struggle with variations in handwriting styles. Deep learning offers a powerful alternative by learning hierarchical features directly from raw pixel data. By integrating a CNN model with a Streamlit web interface, this project enables real-time digit classification and interactive model evaluation.

Objectives:

- To understand and implement a CNN-based OCR system for digit recognition
- To train and evaluate the model using the MNIST dataset
- To deploy the model via a Streamlit interface for real-time predictions and performance visualization

Scope:

The project demonstrates the effectiveness of deep learning in pattern recognition and OCR tasks. It provides a foundation for extending digit recognition to full-text parsing, multilingual OCR, and intelligent document processing systems.

METHODOLOGY

The methodology for this project follows a structured pipeline that combines data preparation, model development, training, and deployment through an interactive web interface. The goal is to build a robust OCR system capable of recognizing handwritten digits with high accuracy and real-time responsiveness.

1. Dataset Description

- Dataset Used: MNIST (Modified National Institute of Standards and Technology)
 - Source: Available via TensorFlow/Keras datasets
 - Composition: 70,000 grayscale images of handwritten digits (0–9), each of size 28×28 pixels
 - Training Set: 60,000 images • Test Set: 10,000 images

2. Data Preprocessing

- Convert uploaded images to grayscale (if not already)
- Normalize pixel values to the range $[0, 1]$
- Reshape images to $28 \times 28 \times 1$ to match CNN input requirements
- One-hot encode labels for training
- Optional augmentation (e.g., rotation, zoom) for robustness

3. Model Architecture

A Convolutional Neural Network (CNN) was designed with the following layers:

- Input Layer: Accepts $28 \times 28 \times 1$ grayscale images
- Convolutional Layers: Extract spatial features using filters
- Activation Functions: ReLU for non-linearity
- MaxPooling Layers: Downsample feature maps
- Dropout Layers: Reduce overfitting
- Fully Connected Layers: Learn high-level representations
- Output Layer: Softmax activation for 10-class digit prediction

4. Training Process

- Loss Function: Categorical Crossentropy
- Optimizer: Adam (adaptive learning rate)
- Batch Size: 32
- Epochs: 10–20 depending on convergence
- Validation: Performance monitored on test set during training

5. Evaluation

- Metrics Used: Accuracy, Precision, Recall, F1 Score
- Visualization Tools: Confusion matrix, per-class accuracy bar chart, training curves
- Streamlit Integration: A “Model Performance” tab displays real-time evaluation results and visual metrics

6. Deployment Interface

- Platform: Streamlit web app
- Features:
 - Upload handwritten digit images for prediction
 - Real-time inference using trained CNN model
 - Interactive visualization of model performance metrics

IMPLEMENTATION AND RESULTS

This project implements a Convolutional Neural Network (CNN) for handwritten digit recognition using Python and TensorFlow/Keras, integrated with a Streamlit-based web interface for real-time predictions and model evaluation. The system allows users to upload digit images and receive instant classification results, while also providing a dedicated “Model Performance” tab for visualizing evaluation metrics.

Implementation Overview:

- Language & Frameworks: Python, TensorFlow/Keras, Streamlit
- Interface: Streamlit web app with upload-and-predict functionality
- Dataset: MNIST — 60,000 training images and 10,000 test images of 28×28 grayscale digits (0–9)

Preprocessing Steps:

- Convert uploaded images to grayscale (if needed)
- Normalize pixel values to the range [0, 1]
- Reshape input to 28×28×1 for compatibility with CNN input layer
- One-hot encode labels for training

Model Architecture:

The CNN model consists of:

- Convolutional Layers: Extract spatial features using filters
- ReLU Activations: Introduce non-linearity
- MaxPooling Layers: Downsample feature maps
- Dropout: Prevent overfitting during training
- Fully Connected Layers: Learn high-level representations
- Output Layer: 10-way Softmax for digit classification (0–9)

Training & Inference:

- Loss Function: Categorical Crossentropy
- Optimizer: Adaptive optimizer (e.g., Adam)
- Batch Size & Epochs: Tuned for convergence and generalization

- Inference: Runs in real time via the Streamlit interface

Evaluation Metrics:

The model was evaluated on the MNIST test set using standard classification metrics. Results are summarized below:

Overall Performance

📊 Accuracy

97.78%

📊 Precision

97.82%

📊 Recall

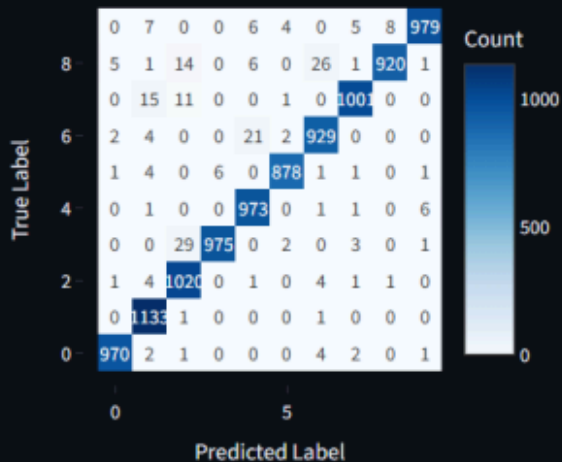
97.78%

📊 F1 Score

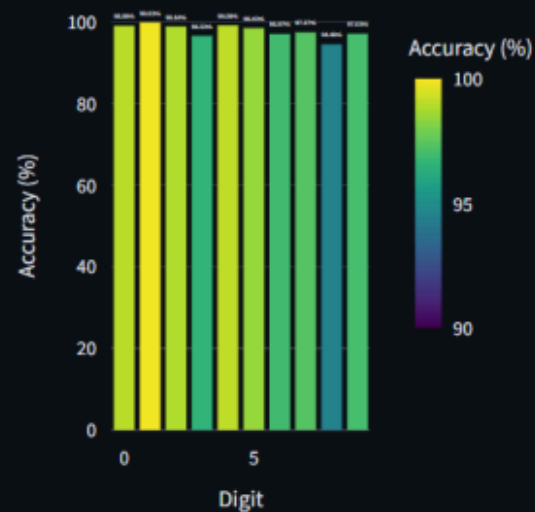
97.78%

Confusion Matrix

Confusion Matrix



Per-Class Accuracy



Detailed Per-Class Metrics

Digit	Correct Predictions	Total Samples	Accuracy (%)
0	970	980	98.98%
1	1133	1135	99.82%
2	1020	1032	98.84%
3	975	1010	96.53%
4	973	982	99.08%
5	878	892	98.43%
6	929	958	96.97%
7	1001	1028	97.37%
8	920	974	94.46%
9	979	1009	97.03%



Summary Statistics

Best Performing Digit: 1
(99.82%)

Most Challenging Digit: 8
(94.46%)

Average Class Accuracy:
97.75%

CONCLUSION

This project successfully demonstrates the use of deep learning for handwritten digit recognition within an OCR framework. The CNN model achieved high accuracy on the MNIST dataset, validating the effectiveness of automatic feature extraction and hierarchical learning.

Key Takeaways:

- Deep learning significantly improves recognition performance over traditional OCR methods.
- CNNs are well-suited for image-based classification tasks due to their spatial feature learning capabilities.

REFERENCES

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3. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, O'Reilly Media, 2022.